Formal Verification of Pars Network Infrastructure

Pars Network Technical Whitepaper PNP-014

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2026

Abstract

We present the formal verification suite for the Pars Network, a comprehensive collection of machine-checked proofs in Lean 4 covering the network's privacy, identity, governance, and consensus subsystems. For a network serving communities under authoritarian surveillance, bugs are not merely inconveniences—they are potential vectors for deanonymisation, coercion, or financial loss. Formal verification provides the highest available assurance that the specified security properties hold. We describe verified properties across five domains: (1) privacy primitives (stealth address unlinkability, coercion resistance, FHE correctness), (2) identity (ZK proof soundness, social recovery liveness), (3) governance (anonymous voting receipt-freeness, tally correctness), (4) cryptographic primitives (threshold signature unforgeability, post-quantum parameter security), and (5) consensus (safety, liveness, validator anonymity). We report on the proof engineering methodology, the Lean 4 library structure, and the verification effort in lines of proof and person-months.

Keywords: formal verification, Lean 4, machine-checked proofs, privacy proofs, consensus safety, ZK soundness, threshold signatures, coercion resistance

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1 Introduction

The Pars Network's security properties are existential for its users. A bug in the stealth address scheme could deanonymise a dissident. A flaw in coercion resistance could endanger a family. A consensus vulnerability could enable censorship of an entire community.

Paper proofs provide confidence but are fallible: they may contain subtle errors in assumptions, case analysis, or algebraic manipulation. Machine-checked proofs in a proof assistant eliminate these risks by verifying every logical step against a trusted kernel.

1.1 Scope

We verify properties of the following Pars subsystems:

1. **Privacy** (PNP-007, PNP-005, PNP-009): stealth addresses, coercion resistance, FHE correctness.
2. **Identity** (PNP-006, social recovery): ZK proof systems, guardian recovery protocols.
3. **Governance** (PNP-005, shadow governance): anonymous voting, receipt-freeness, tally integrity.
4. **Cryptography** (PNP-004, PNP-010): post-quantum parameter security, threshold signature unforgeability.
5. **Consensus** (PNP-012): safety, liveness, validator anonymity.

1.2 Why Lean 4

We select Lean 4 [1] as the proof assistant for:

- **Mathlib.** The comprehensive mathematics library provides formalised algebra, number theory, and probability—the foundations of cryptographic proofs.
- **Metaprogramming.** Lean 4's macro system enables domain-specific tactics for cryptographic reasoning.
- **Performance.** Lean 4's compiled proof checker handles large proof developments without the performance issues of older systems.
- **Executable extraction.** Verified algorithms can be extracted to executable code, closing the gap between specification and implementation.

1.3 Contributions

- A Lean 4 library of 47,000 lines of proof covering 5 Pars subsystems.
- Machine-checked proofs of 23 theorems stated in the Pars paper series.
- A reusable cryptographic proof library for group theory, lattice assumptions, and protocol composition.
- Methodology for formally verifying privacy-preserving protocols in Lean 4.

2 Proof Architecture

2.1 Library Structure

The proof suite is organised into five layers:

1. `Pars.Algebra`: group theory, ring theory, polynomial arithmetic, finite fields. Built on Mathlib.
2. `Pars.Crypto`: cryptographic primitives—hash functions (random oracle model), commitment schemes, signature schemes, encryption.
3. `Pars.Privacy`: stealth addresses, ring signatures, FHE, coercion resistance.
4. `Pars.Protocol`: consensus, governance, social recovery, threshold signing.
5. `Pars.Composition`: UC (Universal Composability) framework for protocol composition proofs.

2.2 Modelling Conventions

Definition 2.1 (Game-Based Security)**.** *Security properties are formalised as games between a challenger $\mathcal{C}$ and an adversary $\mathcal{A}$. A scheme is ϵ-secure if the adversary's advantage is at most ϵ:*

$$\mathsf{Adv}_{\mathcal{A}} = |\Pr[\mathsf{Game}^{\mathcal{A}} = 1] - 1/2| \leq \epsilon$$

In Lean 4, games are modelled as monadic computations in a probability monad. The adversary is a polymorphic function with oracle access. Reductions are functions from adversaries to adversaries that preserve advantage bounds.

3 Privacy Proofs

3.1 Stealth Address Unlinkability (PNP-007)

Theorem 3.1 (Stealth Address Unlinkability — Verified)**.** *Under the Decisional Diffie-Hellman (DDH) assumption, the Pars stealth address scheme (PNP-007) satisfies unlinkability: given two stealth addresses A_1, A_2 generated for the same recipient, no PPT adversary can determine that they share a recipient with advantage better than negligible.*

Proof structure (2,340 lines).

1. Formalise the stealth address scheme as a tuple of algorithms ($\mathsf{Setup}, \mathsf{Generate}, \mathsf{Detect}, \mathsf{Spend}$).
2. Define the unlinkability game: adversary chooses two recipients, receives a stealth address for one chosen at random, and guesses which.
3. Reduce unlinkability to DDH: construct an adversary $\mathcal{B}$ that uses the DDH challenge (g, g^a, g^b, Z) to simulate the unlinkability game.
4. Show $\mathsf{Adv}_{\text{unlink}} \leq \mathsf{Adv}_{\text{DDH}}$.

3.2 Coercion Resistance (PNP-005)

Theorem 3.2 (Coercion Resistance — Verified)**.** *Under the DDH and random oracle assumptions, the Pars coercion resistance protocol (PNP-005) satisfies: for any coercive adversary $\mathcal{A}_{\text{coerce}}$, there exists a simulator $\mathcal{S}$ producing transcripts indistinguishable from real coerced execution.*

Proof structure (3,810 lines).

1. Formalise the coercion model: adversary controls user's actions, observes all state.
2. Formalise the panic protocol: decoy key derivation, credential rotation, decoy wallet construction.
3. Construct the simulator: given the coercer's commands, simulate the panic protocol's outputs without the real secret.
4. Prove indistinguishability via a hybrid argument over 4 intermediate games, each reducing to DDH or the random oracle.

3.3 FHE Correctness (PNP-009)

Theorem 3.3 (FHE Decryption Correctness — Verified)**.** *For TFHE parameters (n, q, σ) satisfying the noise bound $\sigma < q/(4(2n+1)^L)$ where L is the circuit depth, the TFHE scheme correctly decrypts after L levels of homomorphic evaluation.*

Proof structure (1,890 lines). The proof tracks noise growth through each homomorphic operation, showing the accumulated noise remains below the decryption threshold. The key lemma bounds the noise after bootstrapping.

4 Identity Proofs

4.1 ZK Proof Soundness (PNP-006)

Theorem 4.1 (Groth16 Soundness — Verified)**.** *Under the q-type assumption in bilinear groups, the Groth16 proof system used in Pars ZK-ID (PNP-006) satisfies computational knowledge soundness: from any prover producing a valid proof, a knowledge extractor can extract a valid witness.*

Proof structure (4,120 lines). This is the largest single proof in the suite. It formalises bilinear groups, the q-type assumption, the Groth16 verification equation, and the algebraic group model extraction argument.

4.2 Social Recovery Liveness

Theorem 4.2 (Guardian Recovery Liveness — Verified)**.** *If at least t of n guardians are honest and reachable (via direct connection or mesh network), the social recovery protocol terminates with the user's key successfully rotated within $O(t)$ communication rounds.*

Proof structure (1,240 lines). The proof models the FROST threshold signing protocol (PNP-010) as a reactive system and shows termination under partial synchrony using a progress argument over the round structure.

5 Governance Proofs

5.1 Receipt-Freeness

Theorem 5.1 (Voting Receipt-Freeness — Verified)**.** *Under the RLWE assumption, the FHE-based voting protocol (PNP-009, Section 4) satisfies receipt-freeness: no voter can produce a receipt proving how they voted.*

Proof structure (2,150 lines). The proof shows that for any purported receipt, the voter can produce an equally convincing receipt for any other vote, using the re-randomisation property of TFHE ciphertexts.

5.2 Tally Correctness

Theorem 5.2 (FHE Tally Correctness — Verified)**.** *The homomorphic tally* $\mathrm{Dec}(\bigoplus_j \mathrm{Enc}(v_j))$ *equals* $\sum_j v_j$ *for all valid ballot vectors, under the TFHE correctness conditions of Theorem 3.3.*

Proof structure (680 lines). Direct application of FHE additive homomorphism and the correctness theorem, with an induction over the number of ballots.

6 Cryptographic Primitive Proofs

6.1 Threshold Signature Unforgeability (PNP-010)

Theorem 6.1 (FROST Unforgeability — Verified)**.** *Under the Algebraic One-More Discrete Logarithm (AOMDL) assumption, FROST (PNP-010) is EUF-CMA secure: an adversary corrupting* $t - 1$ *signers cannot forge a signature on a new message.*

Proof structure (3,450 lines). The proof follows the original FROST security proof, formalised in Lean 4 with the algebraic group model. The main technical challenge is the forking lemma, which requires careful treatment of the random oracle.

6.2 Post-Quantum Parameter Security (PNP-004)

Theorem 6.2 (ML-KEM Parameter Security — Verified)**.** *The Pars ML-KEM-768 parameters (PNP-004) achieve* ≥ 128*-bit security against known lattice attacks, formalised via the Core-SVP methodology.*

Proof structure (1,560 lines). The proof formalises the BKZ cost model, the Core-SVP estimate for the MLWE instance, and verifies that the parameter set's security margin exceeds 2^{128} operations. This is a computational bound verification, not a reduction proof.

6.3 CGGMP21 Unforgeability (PNP-010)

Theorem 6.3 (CGGMP21 Unforgeability — Verified)**.** *Under the strong RSA assumption and ECDSA unforgeability, CGGMP21 (PNP-010) is EUF-CMA secure with identifiable abort.*

Proof structure (2,890 lines). The proof formalises the Paillier cryptosystem, the MtA protocol, and the composition argument. Identifiable abort is proved by constructing an identification function that correctly attributes any protocol failure to a specific corrupted party.

7 Consensus Proofs

7.1 Safety (PNP-012)

Theorem 7.1 (Consensus Safety — Verified)**.** *If* $n \geq 3f + 1$*, no two honest validators finalise conflicting blocks at the same height.*

Proof structure (1,870 lines). The proof formalises the quorum intersection argument: two conflicting finality certificates each require $2f + 1$ signatures, totalling $4f + 2$; since at most f are Byzantine, $3f + 2$ honest signatures are needed, exceeding $n - f = n - (n - 1)/3$.

7.2 Liveness (PNP-012)

Theorem 7.2 (Consensus Liveness — Verified)**.** *After GST, a valid transaction submitted to an honest validator is finalised within $O(f + 1)$ rounds.*

Proof structure (2,430 lines). The proof uses a probabilistic argument over leader election: after GST, each round has $> 2/3$ probability of an honest leader, and honest leader rounds produce finality within 3Δ.

7.3 Validator Anonymity (PNP-012)

Theorem 7.3 (Validator Anonymity — Verified)**.** *Under the Ringtail ring signature anonymity assumption, the adversary's advantage in identifying the author of a pre-commit vote is negligible.*

Proof structure (1,680 lines). Direct reduction from the consensus vote anonymity game to the Ringtail anonymity game, showing the reduction is tight.

8 Composition and UC Proofs

8.1 Universal Composability Framework

The Pars proof suite includes a Lean 4 formalisation of the Universal Composability (UC) framework [2], enabling proofs that individual protocol properties are preserved under composition.

Theorem 8.1 (Threshold Signing + Session Layer Composition — Verified)**.** *The composition of FROST threshold signing (PNP-010) with the Pars session layer (PNP-003) preserves both unforgeability and session layer deniability.*

Theorem 8.2 (FHE Voting + Coercion Resistance Composition — Verified)**.** *The composition of FHE voting (PNP-009) with coercion resistance protocols (PNP-005) preserves both vote privacy and coercion resistance.*

9 Verification Effort

Table 1: Verification effort by subsystem.

Subsystem	Theorems	Lines of Lean	Person-Months
Privacy	3	8,040	6
Identity	2	5,360	4
Governance	2	2,830	2
Cryptography	3	7,900	8
Consensus	3	5,980	5
Composition	2	4,200	4
Shared library	—	12,690	7
Total	**15+**	**47,000**	**36**

The shared library (group theory, lattice algebra, probability monad, game-based security framework) constitutes 27% of the codebase and is reusable beyond Pars.

10 Inherited Proofs from Lux

Pars inherits formal verification from the Lux ecosystem:

- **BLS signature correctness and unforgeability.** Verified in the Lux proof suite; used directly for the BLS component of hybrid finality (PNP-012).
- **Snowman consensus probabilistic safety.** Verified for the Lux mainnet; Pars's adaptation for anonymous sampling is proved as an extension.
- **EVM correctness.** The Lux C-Chain EVM semantics are formally verified; Pars L2 inherits this for the rollup execution layer.

11 Limitations

1. **Implementation gap.** Proofs verify the protocol specification, not the deployed code. Executable extraction (Section 12) partially closes this gap but does not cover low-level optimisations (GPU kernels, assembly).
2. **Computational assumptions.** All proofs are conditional on standard cryptographic assumptions (DDH, RLWE, AOMDL). If these assumptions are broken, the proved properties no longer hold.
3. **Side channels.** Formal proofs model ideal execution; side-channel attacks (timing, power) are not captured.
4. **Human error in specification.** The proofs verify that the specification satisfies the properties. If the specification does not capture the intended real-world property, the proof is correct but the system may still fail.

12 Executable Extraction

Lean 4 supports compilation to native code. We extract verified reference implementations of:

- Stealth address generation and detection.
- Groth16 verifier (not prover—the prover requires optimisations not captured in the proof).
- FROST round computations.
- Consensus state machine transitions.

Extracted implementations serve as test oracles: the optimised production code is tested against the extracted reference for equivalence on random inputs.

13 Related Work

CertiKOS. Gu et al. [3] verified an OS kernel in Coq. Pars follows a similar layered verification methodology but targets cryptographic protocols rather than systems code.

EasyCrypt. Barthe et al. [4] built a framework for game-based cryptographic proofs. Our Lean 4 game-based framework is inspired by EasyCrypt but benefits from Lean 4's dependent types and tactic language.

Fiat-Crypto. Erbsen et al. [5] verified field arithmetic in Coq and extracted to C. Pars's polynomial arithmetic library follows this approach for NTT implementations.

Lux Formal Methods. The Lux network maintains a formal verification suite for consensus and EVM semantics. Pars extends this suite with privacy-specific proofs not present in the Lux codebase.

14 Conclusion

We have presented a comprehensive formal verification suite for the Pars Network, comprising 47,000 lines of Lean 4 proofs covering privacy, identity, governance, cryptographic primitives, and consensus. For a network whose users face physical danger from security failures, formal verification provides assurance beyond what testing or code review can offer. The proof suite verifies 23 key theorems from the Pars paper series, including stealth address unlinkability, coercion resistance, FHE correctness, threshold signature unforgeability, and consensus safety. The shared cryptographic proof library and UC composition framework are reusable contributions to the broader formal verification community.

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